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RESEARCH INTERESTS

Gamma-ray burst afterglows — structured jets and off-axis geometry; prompt-emission and afterglow modeling; GRB population studies; electromagnetic counterparts of compact-object mergers; fast transients in time-domain surveys and rapid follow-up; image subtraction and machine-learning transient detection.

EDUCATION

Ph.D. in Physics (Astronomy & Astrophysics), Indian Institute of Technology Bombay 2021–2027 (expected)
Study of Electromagnetic Emission from Compact Objects · Advisor: Prof. Varun Bhalerao

M.Sc. Physics, University of Delhi 2018–2020

B.Sc. (Hons.) Physics, Deen Dayal Upadhyaya College, University of Delhi 2015–2018

PUBLICATIONS

22 refereed publications (2 first-author, including one in *Nature*); *h*-index 10. Complete list: [ADS library](#).

First author

1. **Swain, V.**, et al. 2026, *GRB 250704B: An Off-axis Short GRB with a Long-lived Afterglow Plateau*, **ApJL**, 996, L38

A rare case where the afterglow shape is set by viewing geometry rather than energetics or environment: a cosmological analog of GW170817, and a template for interpreting the next gravitational-wave counterpart.

2. **Swain, V.**, et al. 2026, *GRB 230204B: GIT Discovery of a Fast Fading Afterglow Associated with an Energetic Gamma-Ray Burst from a Massive Star Progenitor*, **ApJ**, 1000, 58

Discovery and complete characterization of one of the most energetic bursts known, from prompt gamma-rays through a rapidly fading afterglow to its Wolf–Rayet progenitor wind.

In preparation

3. **Swain, V.**, et al., *The GIT transient-detection pipeline: image subtraction and machine-learning candidate classification* — in preparation, arXiv September 2026
4. **Swain, V.**, et al., *A homogeneously modeled catalog of GIT-observed GRB afterglows* — in preparation

Selected co-author (role in brackets)

- Ho, A. Y. Q., et al. 2023, *Minutes-duration optical flares with supernova luminosities*, **Nature**, 623, 927 [GIT follow-up photometry]
- Kasliwal, M. M., et al. 2025, *AT 2025ulz and S250818k: A Candidate Superkilonova from a Subthreshold Subsolar Gravitational-wave Trigger*, **ApJL**, 995, L59 [GRB afterglow modeling]
- Li, M. L., et al. 2025, *The Nature of Optical Afterglows without Gamma-Ray Bursts: Identification of AT2023lcr*, **ApJ**, 985, 124 [discovered AT2023lcr during ZTFReST scanning]
- Srinivasaragavan, G. P., et al. 2025, *AT 2023sva: a luminous orphan afterglow with evidence for a structured jet*, **MNRAS**, 538, 351 [GIT follow-up]
- Ahumada, T., et al. 2026, *S250206dm: Zwicky Transient Facility Observations*, **PASP**, 138, 034101 [counterpart scanning; GIT follow-up]
- Ahumada, T., et al. 2024, *Searching for Gravitational Wave Optical Counterparts with ZTF: Summary of O4a*, **PASP**, 136, 114201 [counterpart scanning; GIT follow-up]
- Srinivasaragavan, G. P., et al. 2024, *The Broad-lined Ic SN 2023pel from the Energetic GRB 230812B*, **ApJL**, 960, L18 [GIT follow-up; image subtraction across multiple telescopes]
- Eappachen, D., et al. 2026, *Characterizing EP241107a: an Einstein Probe-detected fast X-ray transient*, **MNRAS**, 545 [GIT follow-up; afterglow interpretation]
- Kumar, H., et al. 2022, *India's First Robotic Eye for Time-domain Astrophysics: The GROWTH-India Telescope*, **AJ**, 164, 90 [GIT operations handover; filter color terms]

Plus 11 further refereed publications on supernovae, novae, solar-system objects, and exoplanet ephemerides.

Rapid communications

130 GCN Circulars (20+ first-author, including the GIT afterglow discovery circulars for GRB 230204B and GRB 251126A) · 20 Astronomer's Telegrams · 18 TNS discovery reports as ZTF duty astronomer · minor-planet astrometry in 50+ MPECs.

RESEARCH & OBSERVATIONAL EXPERIENCE

GROWTH-India Telescope (GIT), 0.7 m, Hanle

2021–2026

Core team member for operations and transient follow-up.

- GRB and Einstein Probe follow-up throughout; discovered the optical afterglows of **GRB 230204B** and **GRB 251126A**, the latter with a 4-minute response time (best GIT response: 39 s).
- **Team leader, LVK O4 follow-up operations (2023–2025)** — built and ran the end-to-end response system; triggered on S230615az, S231113bw, and S241125n.
- Near-Earth object follow-up (2021–2024): 100+ objects observed with GIT, contributing to published MPECs.
- **On-site training, Indian Astronomical Observatory, Hanle (2021)** — two-week visit working hands-on with the GIT hardware alongside observatory engineers and observers; also visited the Himalayan Chandra optical and MACE gamma-ray telescopes.

Zwicky Transient Facility — ZTFReST and GROWTH-MMA scanning

- Duty astronomer searching for fast transients; **discovered the orphan afterglow AT2023lcr** (ZTF23aaohpy), published in Li et al. 2025.
- Further discoveries: ZTF23aaxzvr (Galactic nova), ZTF23abbfrqp (flaring candidate); 18 TNS discovery reports.
- ZTF and DECam scanning for optical counterparts of gravitational-wave events and short GRBs.
- Duty astronomer for near-Earth object searches in ZTF data (ZStreaks, 2021–2024).

Target-of-opportunity and DDT observing

Triggered follow-up on HCT, SEDM, NGPS, *Swift*, GMRT, and *Chandra*.

TELESCOPE PROPOSALS

- **PI / Co-I on 21 accepted Himalayan Chandra Telescope proposals** — follow-up of gamma-ray bursts, Einstein Probe transients, and electromagnetic counterparts of gravitational-wave events, including the LIGO IR1 observing cycle.
- **DDT: Chandra** X-ray follow-up of GRB 230812B and GRB 250704B; GMRT radio follow-up of GRB 260516D.

SOFTWARE & PIPELINE DEVELOPMENT

GRB analysis

- End-to-end afterglow analysis pipeline: data ingestion, extinction correction, empirical fitting, physical modeling (nested sampling, evidence/BIC comparison), cross-code systematics validation, degeneracy and parameter-correlation analysis, population statistics, and publication-ready output.
- Prompt-emission pipeline for time-integrated and time-resolved analysis of *Fermi*-GBM and *Swift*-XRT data.

Survey and transient detection

- GIT transient-detection system: source-injection training-set generation, image subtraction, statistically validated analytic vetting, ML-based real/bogus classification, and a human-scanning interface.
- Rebuilt the automated nightly pipeline: reduction, recovery from failed astrometry, saturation and bad-pixel masking, PSF photometry with weight maps, image subtraction, median and RMS-weighted stacking, analytic and ML vetting, and a nightly light-curve interface.

Observatory operations

- Fully automated LVK O4 follow-up system (as team leader): alert and skymap ingestion → tiling →

scheduling → automated or manual triggering → observation and data transfer → reduction → human scanning → GCN publication.

- GIT operations improvements: ToO scheduling algorithm and terminal interface, filter-wheel error handling, weather-alert system, nightly reporting, and multi-institute data synchronization.

TECHNICAL SKILLS

- **Optical/NIR data reduction:** GIT, HCT (optical and NIR), ZTF, LT, SEDM, LDT
- **High-energy data reduction:** *Fermi*-GBM, *Swift*-XRT
- **Modeling frameworks:** *afterglowpy*, *jetsimpy*, *redback*, *VegasAfterglow*, *3ML*, *Fermi GBM Data Tools*, *EXOTIC*
- **Analysis stack:** *Astropy*, *Astroplan*, *Photutils*, *Astro-SCRAPPY*, *SExtractor*, *SEP*, *PSFEx*, *SCAMP*, *SWarp*, *astrometry.net* (*solve-field*)
- **Inference and ML:** *emcee*, *PyMultiNest*, *UltraNest*, *nautilus*, *PyTorch*, *scikit-learn*
- **Computing:** Python, Linux, Git/GitHub, SQL, $\LaTeX$, HTML

AWARDS & HONORS

- **ASI Zubin Kambhavi Award**, Astronomical Society of India, 2026 — team award to the GROWTH-India Telescope team
- **Dhananjay Joshi Fundamental Research Award**, 2026 — team award to the GROWTH-India Telescope team
- Best presentation, live demonstration of the GIT robotic system, Amateur Astronomy Conference, IIT Bombay, 2023
- GATE Scholarship, Government of India, 2021 (doctoral studies); All India Rank 224, GATE Physics 2021; Rank 13, University of Delhi M.Sc. entrance examination

TALKS & PRESENTATIONS

Invited

- *A Treasure Hunt in the Sky: Hands-on Exploration of Optical and NIR Archival Surveys* — public talk, Wilson College (Autonomous), Mumbai 2025

Contributed talks

- *GRB 250704B: An Off-axis Short GRB with a Long-lived Afterglow Plateau* — Young Astronomers' Meet 2025
- *GRB 250704B* — SymPhy national conference, IIT Bombay 2025
- *GRB 230204B: A Highly Energetic and Fast Fading GRB* — 42nd Astronomical Society of India (ASI) Meeting 2024
- Live demonstration of the GIT robotic system — Amateur Astronomy Conference, IIT Bombay 2023
- Regular talks at the IITB Astronomy Journal Club and the Time-Domain Astronomy Club (IIT Bombay, TIFR, IIA, IUCAA)

Posters

- *Rapid follow-up of gravitational wave events with a robotic meter-class optical telescope* — IAU General Assembly XXXII, Cape Town (2024); 43rd ASI Meeting 2025
- *Studying Near-Earth Asteroids with the wide-field GROWTH-India Telescope* — 41st ASI Meeting 2023
- *Solar System Studies with the GROWTH-India Telescope* — 40th ASI Meeting and Young Astronomers' Meet 2022

TEACHING & MENTORING

Teaching assistant, IIT Bombay — Electronics Laboratory (2022–23), Astronomy & Astrophysics (2023–24), Undergraduate Physics Laboratory (2024–25)

Research mentoring — supervised 20+ undergraduate and junior graduate students (2021–present) on projects spanning telescope operations, pipeline development, and GRB science:

- *Observations and operations*: GIT observing and robotic operations, automated focus loops, new camera installation and automation, dashboard and database interfaces — S. Samantaray, M. Gulati, K. Adithya Krishna, R. Sharma, Ravi Kumar, T. Mohan
- *Transient detection*: image subtraction, source injection for training-set generation, ML real/bogus classification, scanning interfaces — A. Suresh, D. Raman, S. Gholap, M. Goyal, A. Lonakadi, V. Vijaykumar
- *GRB science*: multiband and broadband afterglow modeling, catalog data assembly, *Fermi*-GBM prompt analysis — Y. Wagh, M. Goyal, Aryan Kumar, H. Tanenia, Athul, S. Patil
- *Gravitational-wave follow-up*: GIT tiling of LVK skymaps — H. Choudhary, T. Roychowdhury
- *Solar system*: near-Earth object observing, ZTF streak scanning, Astreaks pipeline development — H. Choudhary, Y. Wagh, S. Tiwari, V. Vijaykumar
- Several mentees are now first authors on GIT publications and circulars; two are continuing my GIT operations and afterglow-modeling work.

Krittika Summer Projects 5.0 (two-month program, 2024) — *Study of the Nearest Hydrogen-Rich Core-Collapse Supernova: SN 2023ixf*

Outreach mentoring — supervised 9 school and first-year students (2023–2026) on short research projects in GRB physics and optical astronomy.

COLLABORATIONS

GROWTH-India — operations, observing, and pipeline development · **ZTF (ZTFReST)** — fast-transient scanning · **GROWTH-MMA** — ZTF and DECam scanning for gravitational-wave and short-GRB counterparts · **LSST-FAST** — fast-transient searches with Vera C. Rubin Observatory

SERVICE & OUTREACH

- Volunteer, *Transients 2024* conference and the SERB National Research meeting, IIT Bombay
- Public demonstrations of LIGO, EMGW, and Daksha satellite science at IIT Bombay Techfest (2021–2026) and VJTI Mumbai (2023)
- Regular live demonstrations of GIT operations for laboratory visitors
- Volunteer educator, Padhao Campaign (Rann Bhoomi Foundation) and NSS, University of Delhi (2016–17)

REFERENCES

- **Prof. Varun Bhalerao** — IIT Bombay (PhD advisor) — varunb@iitb.ac.in